CS675
Computer Graphics

2D Transformations

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VIGIL
VISION, GRAPHICS, IMAGING & LEARNING

Transformations

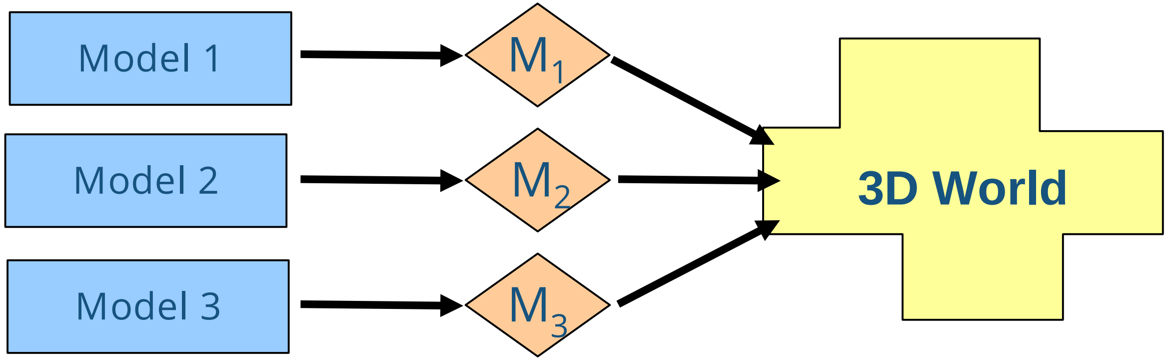
- What is a transformation?
  - It is a map from one vector space to another i.e., it maps the vectors of one vector space to the vectors of another vector space.
- Why is it useful?
  - Modelling
    - Specify object position, orientation, size in the world.
    - Create multiple instances of template shapes.
    - Specify hierarchical models.
  - Viewing
    - Makes viewing window and device independent
    - Synthetic Camera Model

The Modelling-Viewing pipeline

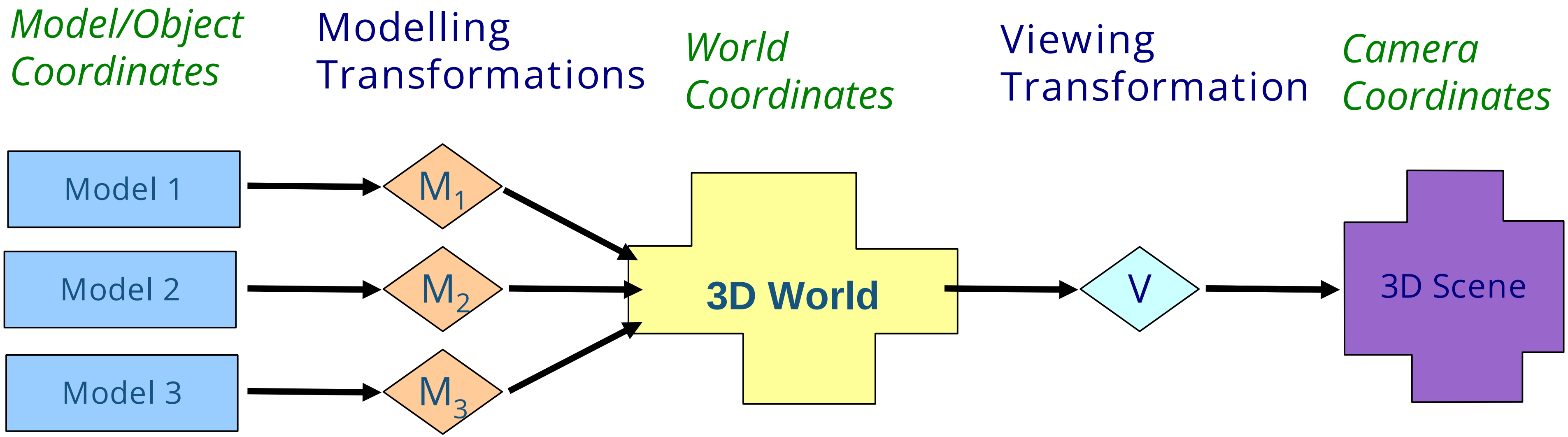
Model/Object Coordinates

Modelling Transformations

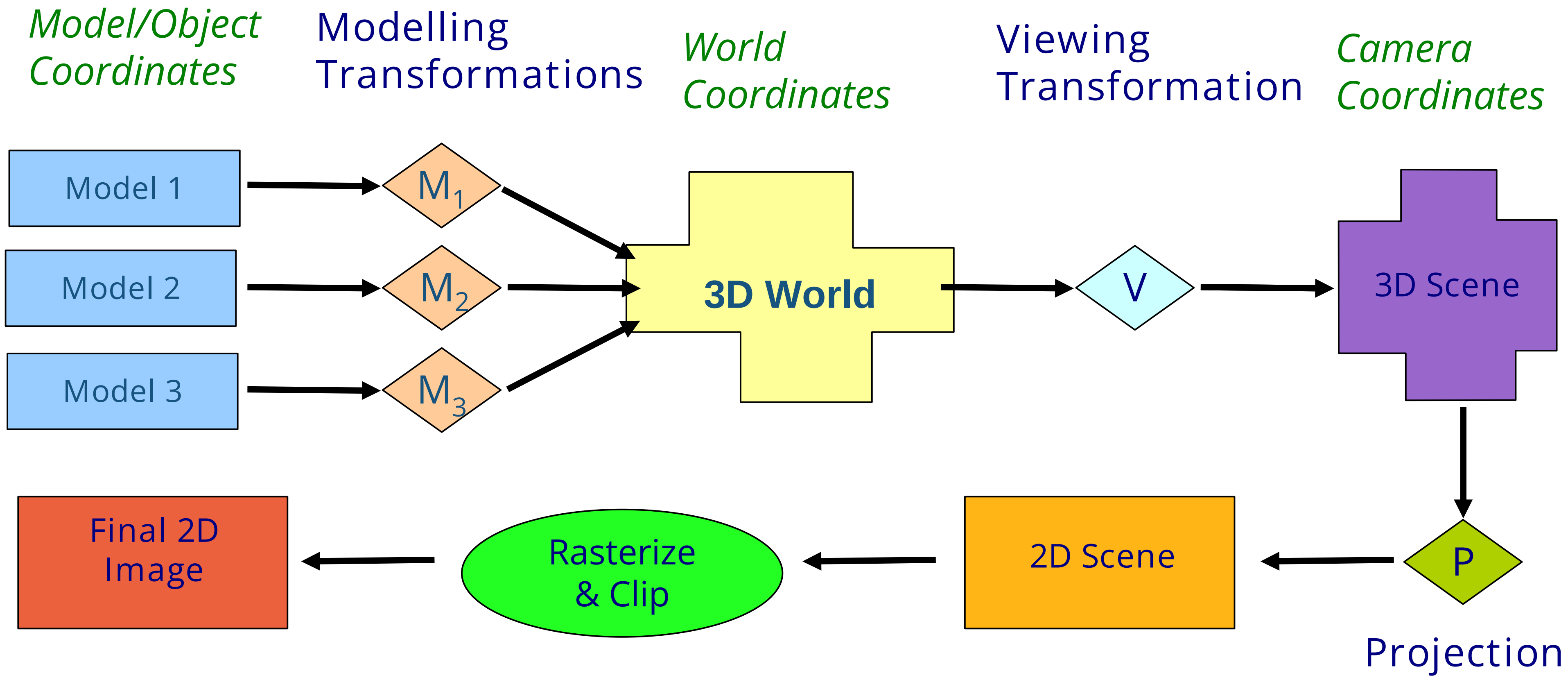
World Coordinates



The Modelling-Viewing pipeline



The Modelling-Viewing pipeline



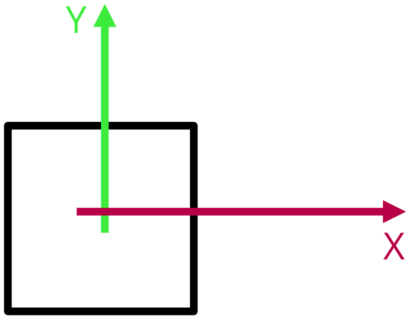
Notation

- We will represent a vector $U = u_x\hat{\imath} + u_y\hat{\jmath} + u_z\hat{k}$ as a column vector $U = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix}$
- We represent transformations as matrices.
- We will apply the transformation to the vector by pre-multiplying the vector with the transformation.

$$U' = \begin{bmatrix} u'_x \\ u'_y \\ u'_z \end{bmatrix} = TU = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix}$$

2D Transformations - Scaling

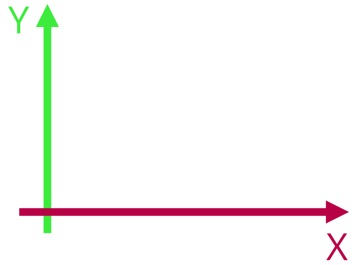
$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \qquad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$



$$P' = SP$$

$$p'_x = s_x . p_x$$
$$p'_y = s_y . p_y$$

$$\begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$



2D Transformations - Scaling

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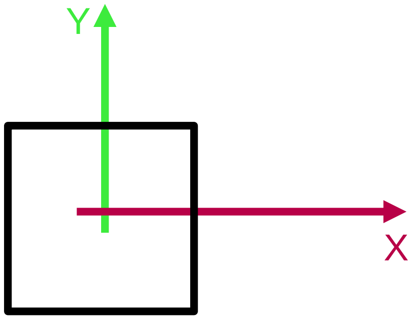

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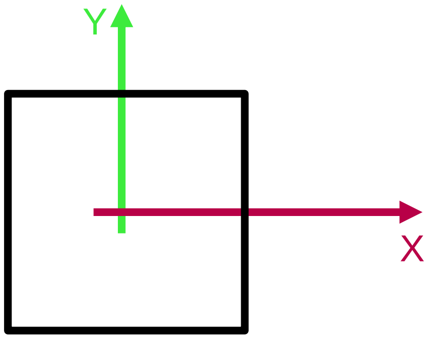
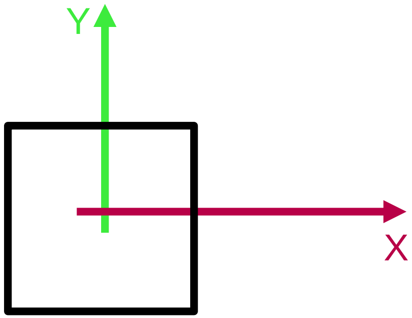


Uniform or Isotropic Scaling $\qquad s_x = s_y$

2D Transformations - Scaling

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \qquad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$



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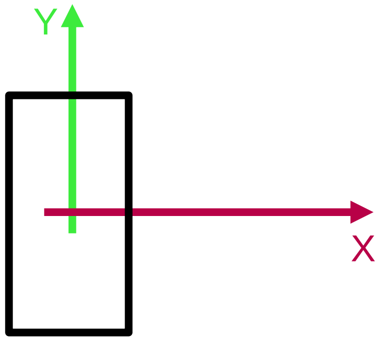


Nonuniform or Anisotropic Scaling $\qquad s_x \neq s_y$

2D Transformations - Rotation

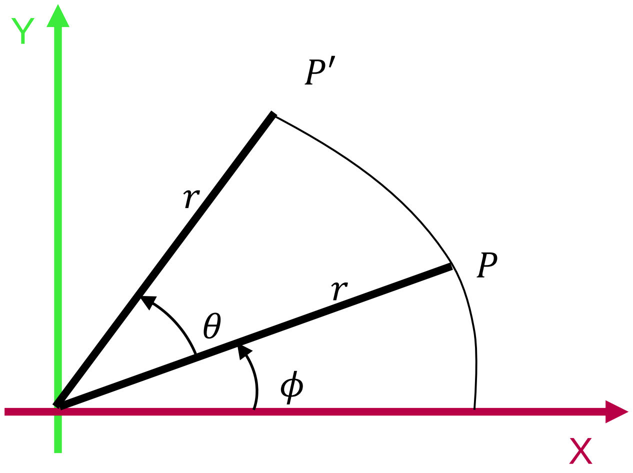
$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} = \begin{bmatrix} rcos(\phi) \\ rsin(\phi) \end{bmatrix}$$

$$P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} rcos(\phi + \theta) \\ rsin(\phi + \theta) \end{bmatrix} = \begin{bmatrix} rcos(\phi)cos(\theta) - rsin(\phi)sin(\theta) \\ rcos(\phi)sin(\theta) + rsin(\phi)cos(\theta) \end{bmatrix}$$

$$P' = R.P$$

$$\begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{bmatrix} . \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$R = R_\theta = \begin{bmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{bmatrix}$$



2D Transformations

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$P' = TP \;\Rightarrow\; \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} . \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = ap_x + cp_y$$
$$p'_y = bp_x + dp_y$$

For $a = 1, d = 1$ and $b = 0, c = 0$

$i.e., T = I$ (Identity Transformation)

$$p'_x = p_x$$
$$p'_y = p_y$$

2D Transformations

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$P' = TP \;\Rightarrow\; \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} . \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = ap_x + cp_y$$
$$p'_y = bp_x + dp_y$$

For $b = 0, c = 0$

$i.e., T = S$ (Scaling Transformation)

$$p'_x = a.p_x$$
$$p'_y = d.p_y$$

2D Transformations

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$P' = TP \;\Rightarrow\; \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \cdot \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = ap_x + cp_y$$
$$p'_y = bp_x + dp_y$$

For $a = -1, d = 1$ and $b = 0, c = 0$

$i.e., T = Rf$ (Reflection Transformation)

$p'_x = -p_x$ Reflection about the line

$p'_y = p_y$ $x = 0$

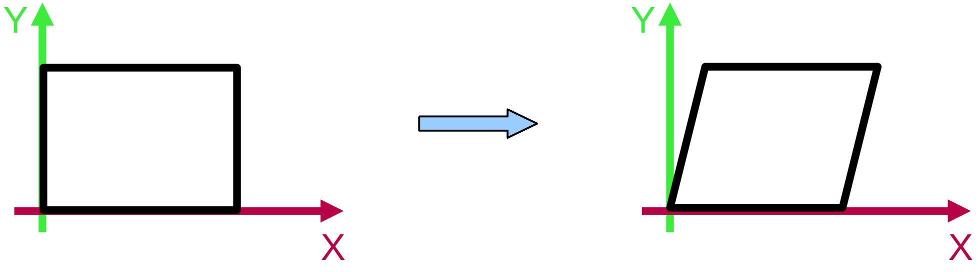
2D Transformations - Shear

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad T = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$$

$$P' = TP \;\Rightarrow\; \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = p_x + c p_y$$
$$p'_y = p_y$$

Shearing in X



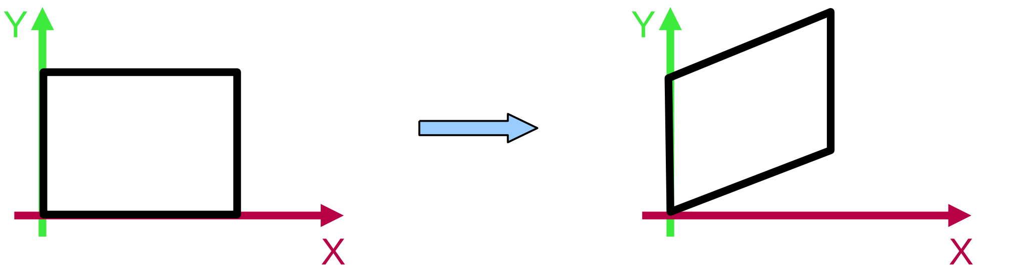
2D Transformations - Shear

$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \qquad T = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}$$

$$P' = TP \;\Rightarrow\; \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} . \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = p_x$$
$$p'_y = bp_x + p_y$$

Shearing in Y



2D Transformations - Translation

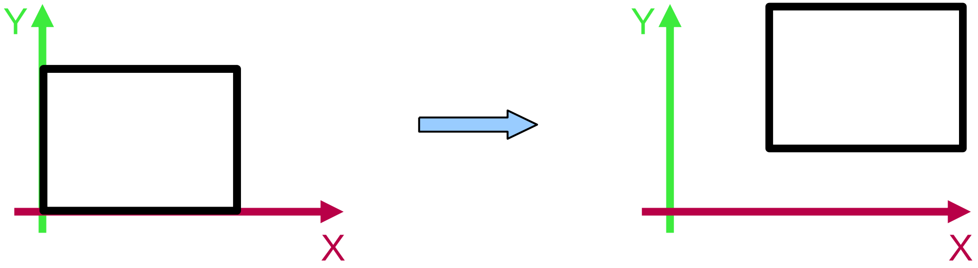
$$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \quad P' = \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \quad T = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$P' = T + P \Rightarrow \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} + \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$p'_x = t_x + p_x$$
$$p'_y = t_y + p_y$$

Different from other transformations!

Cannot represent it as a matrix multiplication.



Linear Transformations

Given two vectors $u, v \in V$, from a vector space V and a transformation $T: V \to W$, where W is also a *vector space.*

The transformation T is a *linear* map iff

$$T(u + v) = Tu + Tv$$
$$T(\alpha u) = \alpha\, (Tu)$$

where α is a scalar.

Linear Transformations

Prove the following

- 2D Scaling, Rotation, Reflection, Shear are all linear transformations.
- 2D Translation is not a linear transformation.

Homogenous Coordinates

$P = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$ in homogenous coordinates becomes $P_h = \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix}$

A general point in homogenous coordinates can be mapped back to usual non-homogenous coordinates as follows by dividing with the homogenous dimension.

$$P_h = \begin{bmatrix} wp_x \\ wp_y \\ w \end{bmatrix} \sim P = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

Therefore, the homogenous points $\begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 12 \\ 8 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix},$

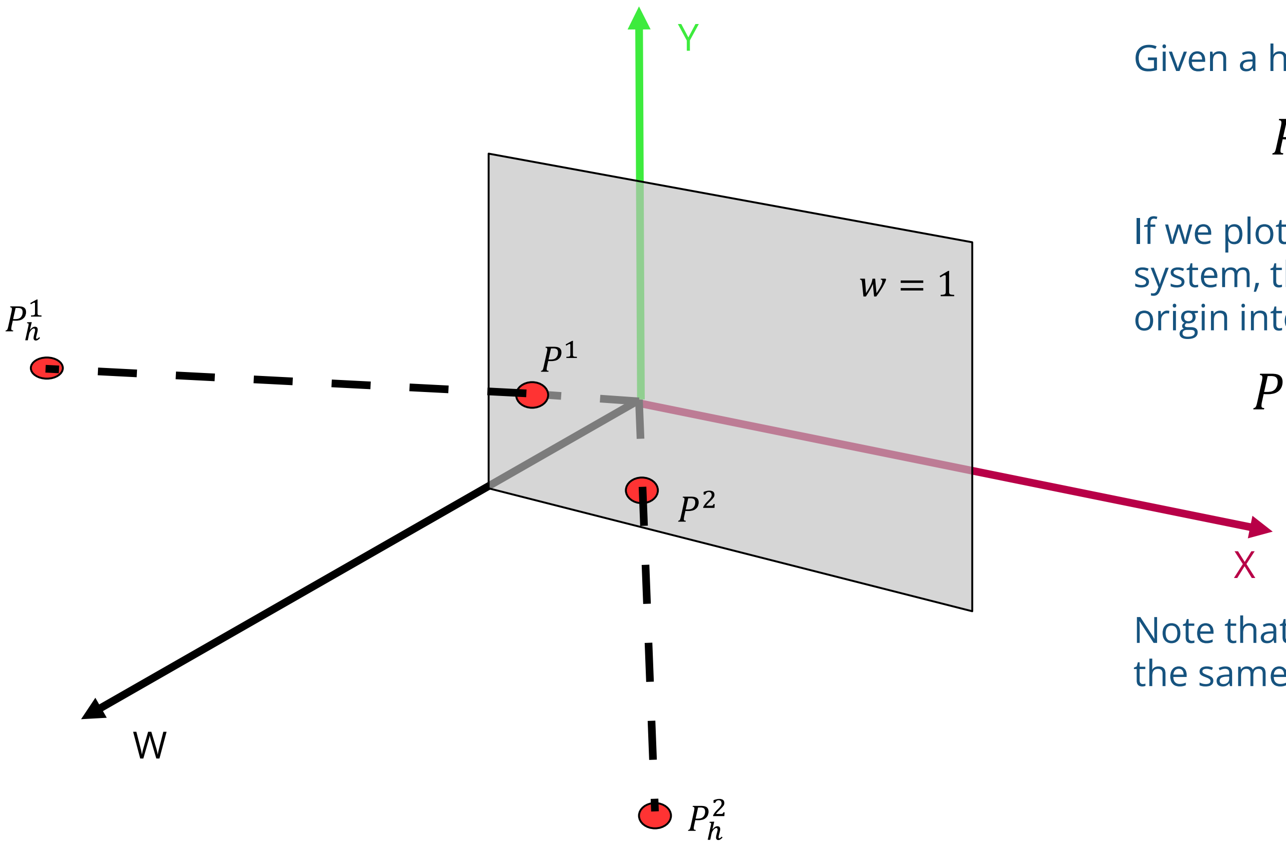
all represent the same 2D point $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Homogenous Coordinates

Another name for the usual 2D non-homogenous coordinates is 2D Euclidean coordinates. We represent these as $\mathbb{R}^2$

Another name for the corresponding 3D homogenous coordinates is 3D projective coordinates. We represent these as $\mathbb{P}^3$

Homogenous Coordinates



Given a homogenous point:

$$P_h = [x \quad y \quad w]^T$$

If we plot the point in the XYH coordinate system, then line joining the point with the origin intersects the $w = 1$ plane at:

$$P = [x/w \quad y/w \quad 1]^T$$

Note that all points on the line will map to the same point on the $w = 1$ plane.

2D Transformations

The general 2D Transformation matrix now becomes 3x3: $\begin{bmatrix} a & c & l \\ b & d & m \\ 0 & 0 & 1 \end{bmatrix}$

i.e.,

$$\begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \begin{bmatrix} a & c & l \\ b & d & m \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = ax + cy + l$$

$$y' = bx + dy + m$$

$$w = 1$$

2D Transformations

The general 2D Transformation matrix now becomes 3x3: $\begin{bmatrix} a & c & l \\ b & d & m \\ 0 & 0 & 1 \end{bmatrix}$

i.e.,

$$\begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \begin{bmatrix} a & c & l \\ b & d & m \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = ax + cy + l$$
$$y' = bx + dy + m$$
$$w = 1$$

This is known as an *Affine* Transformation.

2D Transformations

So we see that for translating a point we use the matrix: $\begin{bmatrix} 1 & 0 & l \\ 0 & 1 & m \\ 0 & 0 & 1 \end{bmatrix}$

Therefore the new transformed points become:

$$\begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & l \\ 0 & 1 & m \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = x + l$$
$$y' = y + m$$

Scaling/Rotation/Shear continue to work as before using the matrix: $\begin{bmatrix} a & c & 0 \\ b & d & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Concatenating Transformations

$$T_1 = \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & m_1 \\ 0 & 0 & 1 \end{bmatrix} \qquad T_2 = \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & m_2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = T_1 . P = \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & m_1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad P'' = \begin{bmatrix} x'' \\ y'' \\ w'' \end{bmatrix} = T_2 . P' = \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & m_2 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix}$$

Successive Translations are *additive*.

$$P'' = T_2 . T_1 . P = \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & m_1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & m_2 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & l_1 + l_2 \\ 0 & 1 & m_1 + m_2 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Concatenating Transformations

$$S_1 = \begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad S_2 = \begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = S_1 . P = \begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad P'' = \begin{bmatrix} x'' \\ y'' \\ w'' \end{bmatrix} = S_2 . P' = \begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix}$$

Successive Scalings are *multiplicative*.

$$P'' = S_2 . S_1 . P = \begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_{x1} . s_{x2} & 0 & 0 \\ 0 & s_{y1} . s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Concatenating Transformations

$$R_\theta = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R_\phi = \begin{bmatrix} cos\phi & -sin\phi & 0 \\ sin\phi & cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = R_\theta . P = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$P'' = \begin{bmatrix} x'' \\ y'' \\ w'' \end{bmatrix} = R_\phi . P' = \begin{bmatrix} cos\phi & -sin\phi & 0 \\ sin\phi & cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix}$$

Concatenating Transformations

Successive Rotations are *additive*.

$$P'' = R_\phi . R_\theta . P = \begin{bmatrix} cos\phi & -sin\phi & 0 \\ sin\phi & cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

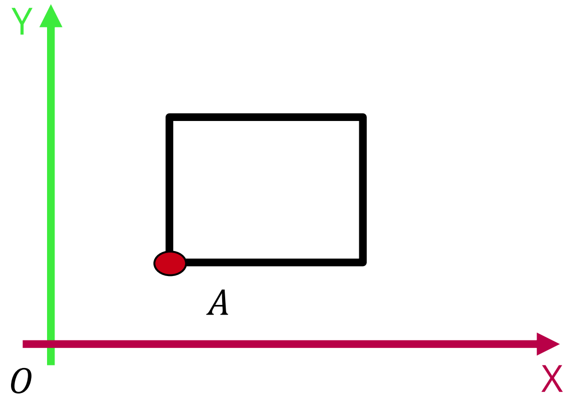
$$P'' = \begin{bmatrix} cos(\phi+\theta) & -sin(\phi+\theta) & 0 \\ sin(\phi+\theta) & cos(\phi+\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Concatenating Transformations

Rotation about an arbitrary point A

We know how to rotate about the origin O

- Translate A to O
- Rotate about O
- Translate back to A



Concatenating Transformations

Rotation about an arbitrary point A

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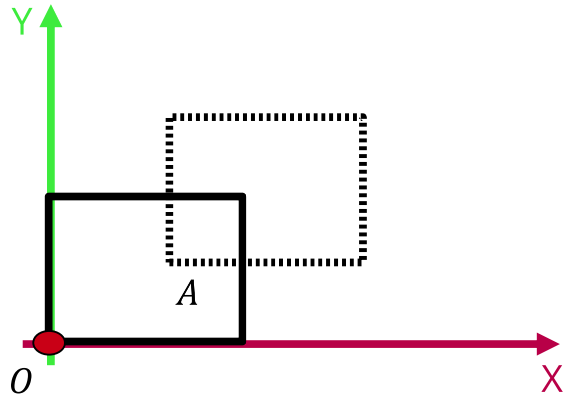


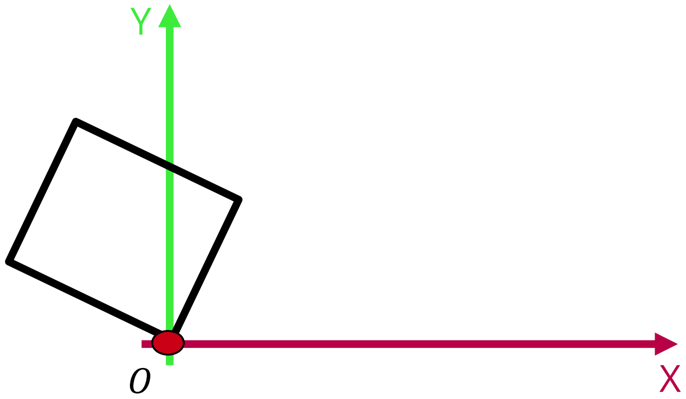
- Translate A to O

$$P' = \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = T_1 . P = \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & m_1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Concatenating Transformations

Rotation about an arbitrary point A

We know how to rotate about the origin O



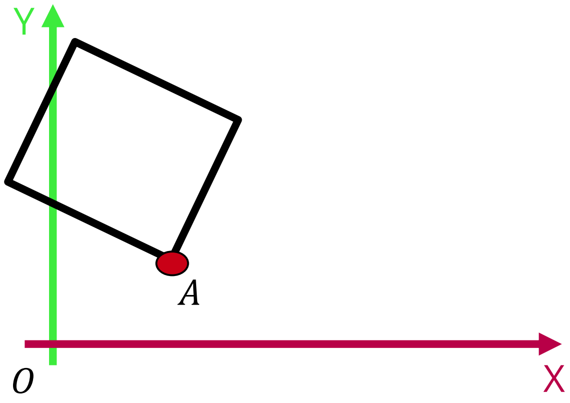
- Translate A to O
- Rotate about O
- Translate back to A

$$P'' = \begin{bmatrix} x'' \\ y'' \\ w'' \end{bmatrix} = R_\theta . P' = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix}$$

Concatenating - Transformations

Rotation about an arbitrary point A

We know how to rotate about the origin O



- Translate A to O
- Rotate about O
- Translate back to A

$$P''' = \begin{bmatrix} x''' \\ y''' \\ w''' \end{bmatrix} = T_2 . P'' = \begin{bmatrix} 1 & 0 & -l \\ 0 & 1 & -m \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x'' \\ y'' \\ w'' \end{bmatrix}$$

Concatenating Transformations

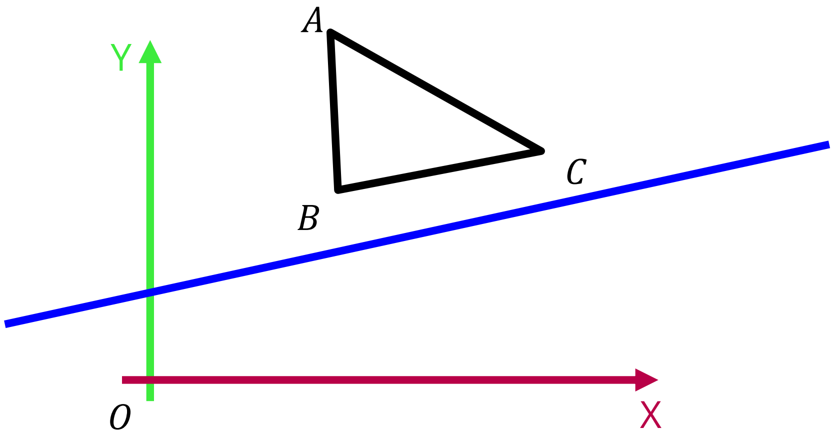
The composite transformation is then:

$$T_2.R_\theta.T_1 = \begin{bmatrix} 1 & 0 & -l \\ 0 & 1 & -m \\ 0 & 0 & 1 \end{bmatrix}.\begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.\begin{bmatrix} 1 & 0 & l \\ 0 & 1 & m \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} cos\theta & -sin\theta & lcos\theta - msin\theta - l \\ sin\theta & cos\theta & lsin\theta + mcos\theta - m \\ 0 & 0 & 1 \end{bmatrix}$$

Concatenating Transformations

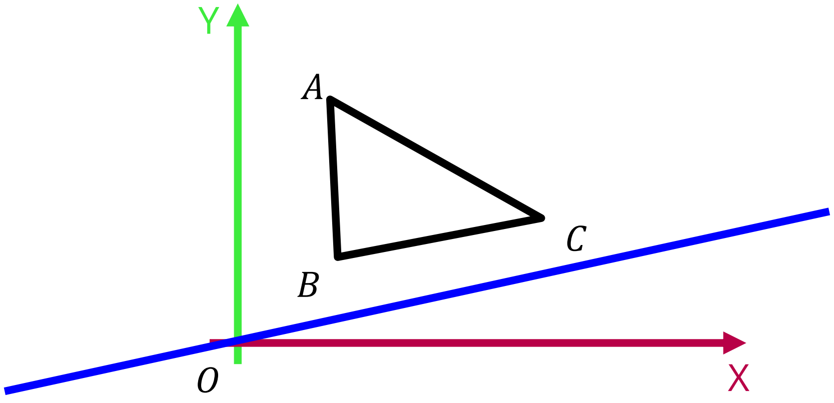
Reflection about an arbitrary line.



Concatenating Transformations

Reflection about an arbitrary line.

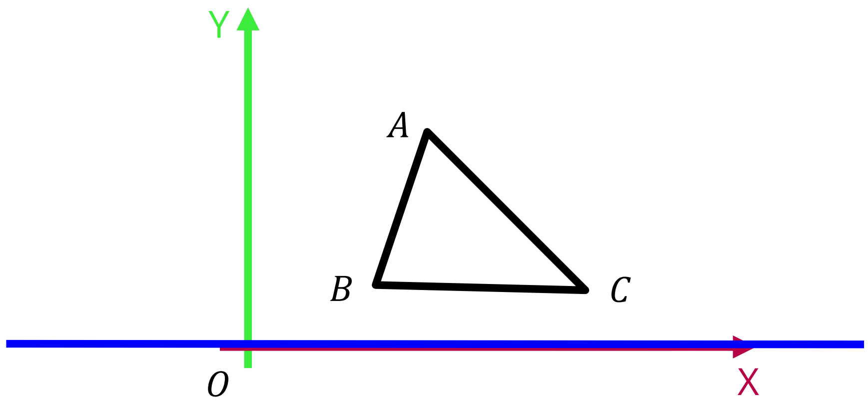
- Translation



Concatenating Transformations

Reflection about an arbitrary line.

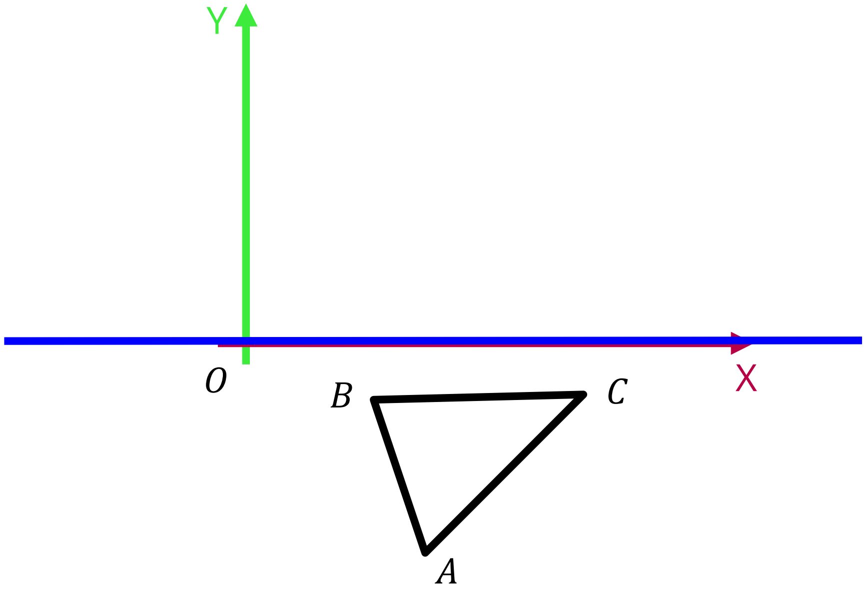
- Translation
- Rotation



Concatenating Transformations

Reflection about an arbitrary line.

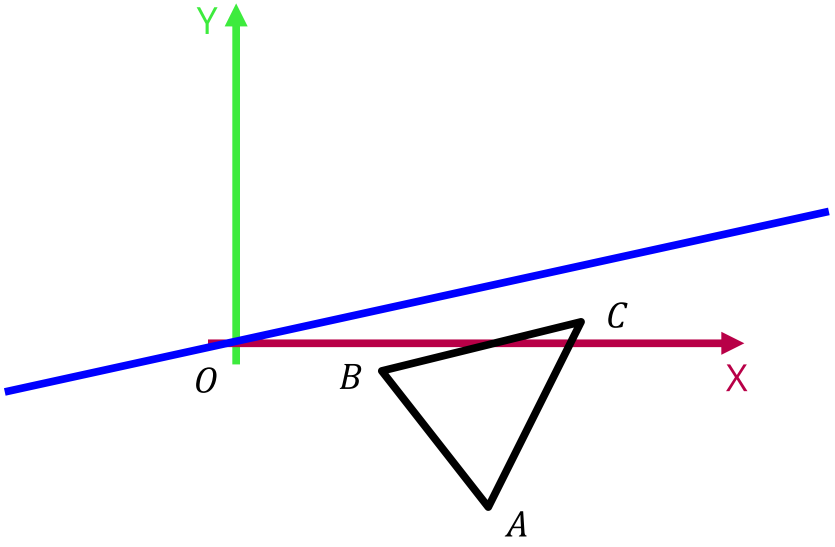
- Translation
- Rotation
- Reflection



Concatenating Transformations

Reflection about an arbitrary line.

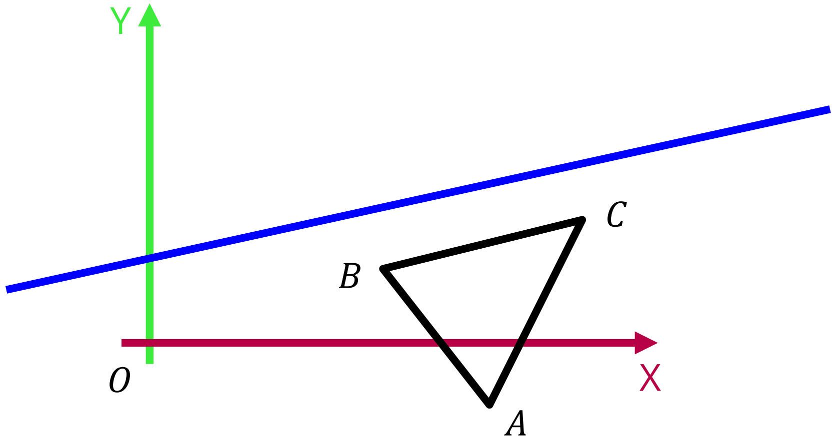
- Translation
- Rotation
- Reflection
- Rotation



Concatenating Transformations

Reflection about an arbitrary line.

- Translation
- Rotation
- Reflection
- Rotation
- Translation



Concatenating Transformations

In general, transformation composition is not commutative.

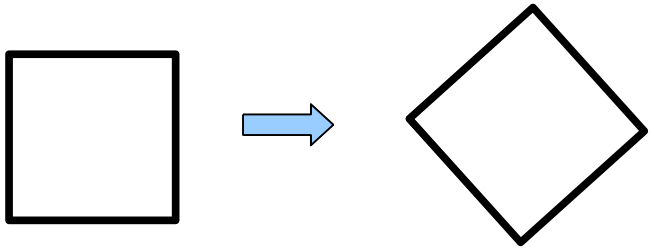
$$T_1 . T_2 \neq T_2 . T_1$$

2D Transformations

Rigid Transformations

- A square remains a square.
- Preserves lengths and angles.
- Rotations and Translations

$$T = \begin{bmatrix} r_{11} & r_{12} & l \\ r_{21} & r_{22} & m \\ 0 & 0 & 1 \end{bmatrix}$$

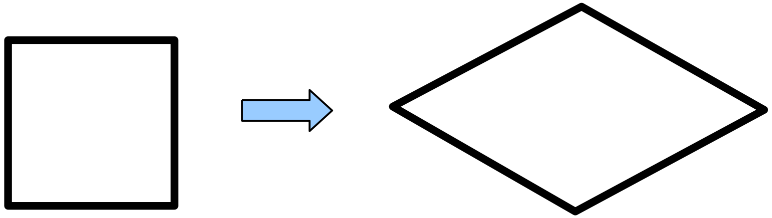


2D Transformations

Affine Transformations

- Preserves parallelism.
- Rotations, Translations, Scaling and Shears.

$$T = \begin{bmatrix} a & c & l \\ b & d & m \\ 0 & 0 & 1 \end{bmatrix}$$



2D Transformations

General 2D Transformation

$$T = \left[\begin{array}{cc|c} a & c & l \\ b & d & m \\ \hline p & q & s \end{array}\right]$$